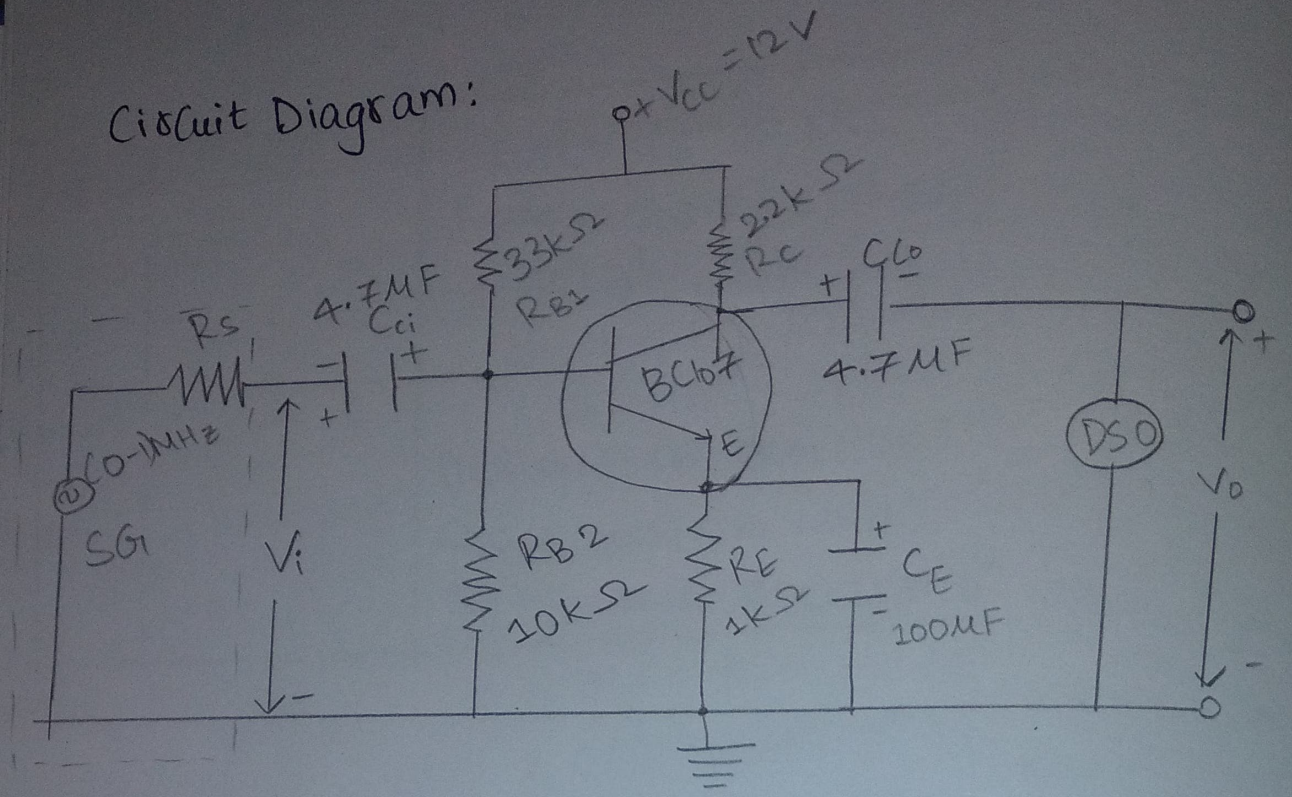
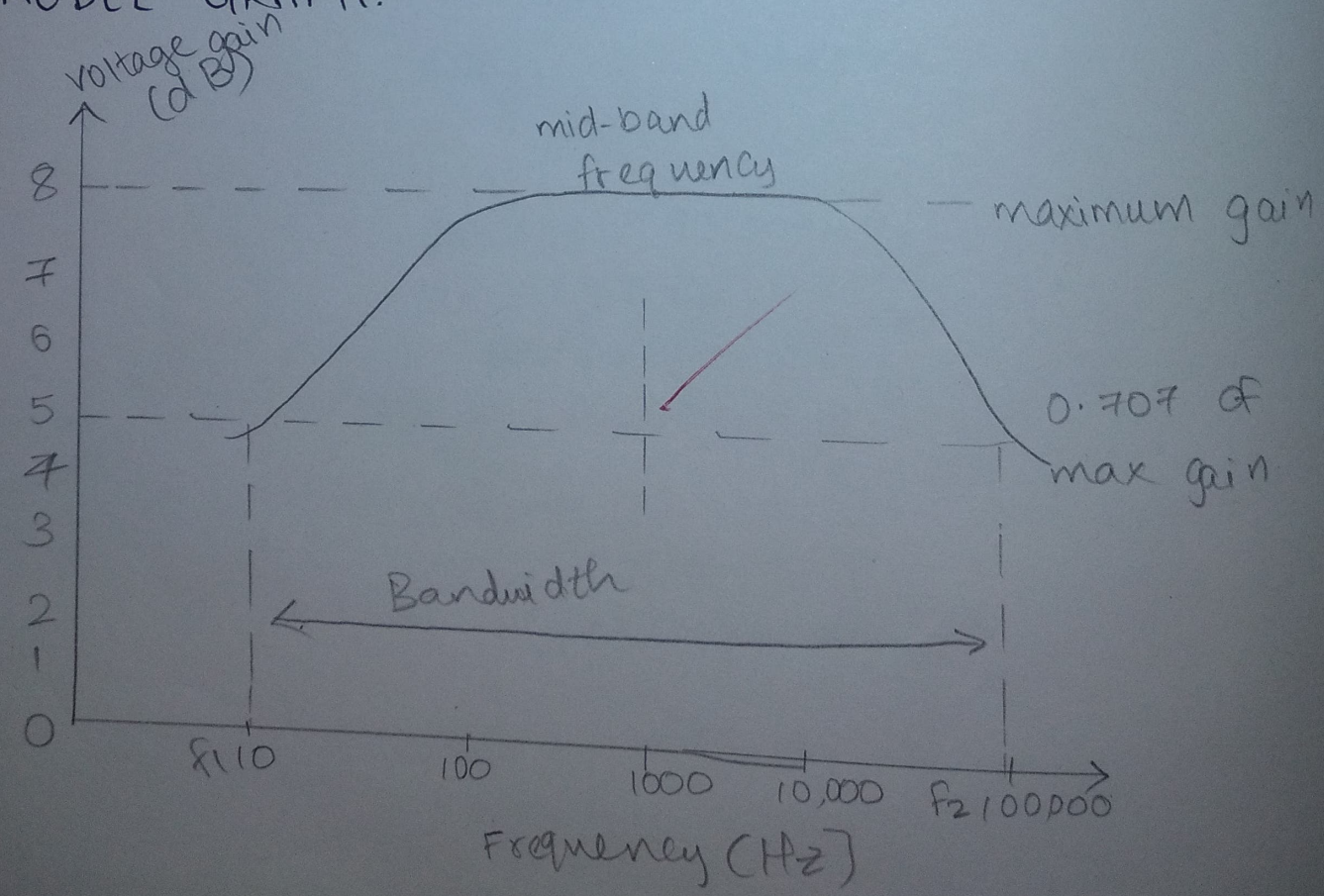
Circuit Diagram:



MODEL GRAPH:



EXP NO: 1 DATE:

FREQUENCY RESPONSE OF COMMON EMITTER AMPLIFIER

AIM:

To design a single-stage common-emitter amplifier using a transistor, simulate the circuit in LTspice, analyse its frequency response, and calculate gain and bandwidth.

SOFTWARE REQUIRED:

LTspice simulator.

PROCEDURE:

- Connect the circuit as per the circuit diagram using LTspice simulator.

DC Analysis:

- Set Vin = 0 by reducing the amplitude of the input signal from signal generator.
- Open circuit the capacitors, since it block DC voltage

Vin = 1V

S.NO	FREQUENCY (Hz)	V_o (V)	V_o/V_{in} (V)	Gain = 20log (V_o/ V_{in}) (dB)
1	10	0.5	0.5	-6.02
2	50	0.7	0.7	-3.10
3	100	1.0	1.0	0
4	1K	2.5	2.5	7.96
5	10K	2.8	2.8	8.94
6	50K	2.7	2.7	8.63
7	100K	2.0	2.0	6.02
8	200K	1.2	1.2	1.58

Calculation:

1. Gain Calculation:

$$Gain(dB) = 20\log\left(\frac{V_o}{V_{in}}\right)$$

$$V_{in} = 1V$$

$$Gain(dB) = 20\log V_o$$

For frequency = 1kHz

$$V_o = 2.5V$$

$$Gain = 20\log V_o = 20\log(2.5)$$

$$= 20 \times 0.397 = 7.69dB$$

For frequency = 10Hz

$$V_o = 0.5V$$

$$Gain = 20\log V_o = 20\log(0.5)$$

$$= 20 \times (-0.301)$$

$$= -6.02dB$$

Bandwidth:

$$Bandwidth = f_H - f_L$$

$$f_H = 100kHz, \quad f_L = 50Hz$$

$$BW = 100000 - 50 = 99950 \simeq 100kHz$$

RESULT:

The common Emitter Amplifier was constructed and the following results were observed.

1. Maximum Signal handling capacity 2.8 V
2. Q-point : I_{CQ} mA
   V_{CEQ} 6 V
3. Gain of the amplifier 8.9 dB
4. Bandwidth of the amplifier 100kHz
5. Gain-Bandwidth Product: 8×10^5